

4(a)	<i>any four from</i> I signals R nerve impulse first time <i>SAN</i> 1 acts as a pacemaker / initiates heart beat / initiates cardiac cycle ; A regulates heartbeat A described, e.g. as rhythm / emits impulses at regular intervals 2 releases / AW, waves of excitation / (electrical) impulses ; A ref. to, action potentials / depolarisation R nerve impulses 3 spread across / AW, atria / atrial walls or leads to atrial, systole / contraction(s) ; <i>AVN</i> 4 allows a (short) delay / ~0.1 s ; 5 passes the impulse / wave of excitation, to the Purkyne fibres / down the septum ; A Bundle of His R nerve impulse 6 detail ; - e.g. so atria contract before ventricles - allows ventricles to fill / allow atria to empty completely - so atria have, emptied / contracted, before ventricular contraction begins - so atria and ventricles don't contract at the same time	4
4(b)	<i>any two from</i> - contraction / ventricular systole, begins at the base of the <u>ventricles</u> ; - both ventricles contract at the same time ; - blood is forced, upwards / into arteries / through (named) semi-lunar valves ; A pushed <i>idea that</i> so ventricles, empty / pump out most of the blood ; - AVP ; - e.g. so impulses / waves of excitation, travel upwards from the base	2

Question	Answer	Marks
4(c)	<i>any four from</i> 1 noradrenaline is, signalling, molecule / compound ; A noradrenaline is the signal I noradrenaline is a (neuro)transmitter 2 SAN cell is the <u>target</u> cell ; 3 noradrenaline / signalling molecule, binds to (cell surface) receptors (of target cell) ; A 'fits' R 'receptor cells' 4 receptors are specific to noradrenaline ; A <i>ref.</i> to complementary (shapes) 5 <i>ref.</i> to formation of cAMP / second messenger <i>or</i> activating enzyme ; 6 response is the opening of channel proteins (so calcium ions enter) ;	4